

Homework-2

Question 1

```
install.packages("aRtsy")
```

install.packages installs the package to your device, whereas the library function tells R to use the specific package in the code below the library function

Packages

Question 2

```
library(aRtsy)
```

The Collatz conjecture is a famous unsolved math problem that asks whether two simple arithmetic operations will eventually transform every positive integer into 1. If the number is even, you divide by 2, and if the number is odd, you multiple by 3 and add 1.

$$f(x) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ 3n + 1 & \text{if } n \text{ is odd} \end{cases}$$

Question 3

```
canvas_collatz(  
  colors = c("blue"),  
  n = 10  
)
```

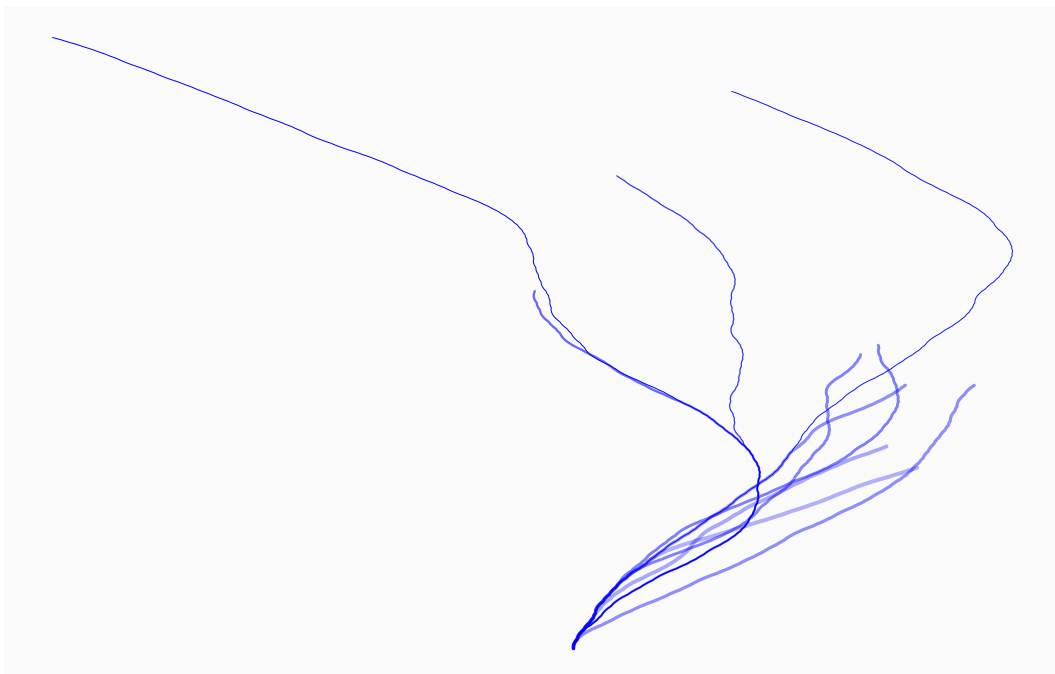
The above code uses the `canvas_collatz` function from the `artsy` package. It generates artwork based on the Collatz conjecture by calculating the starting paths for 10 randomly chosen starting integers. It plots these fractal paths on a canvas using the designated color.

part b

You get a different output every time because the code runs based off of 10 randomly generated integers. Since these 10 numbers change each time, every time you render this into a quarto file, it would show a different graph.

By setting the seed through the `set.seed` function, we can choose how the random numbers are generated through the `canvas_collatz` function.

```
set.seed(329386292)  
  
canvas_collatz(  
  colors = c("blue"),  
  n = 10)
```



Question 4